

Green Score — algorithms, sources, and an audit

From: Engineering
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Subject: What the Green Score computes, where every number comes from, and how accurate it is

Summary

This sets out the Green Score end to end: the algorithm, where every constant comes from, what the model actually produces, and — new in this revision — **how accurate it is against satellite measurement**, in both senses: how close the ward-level temperature is (§5A), and whether the pattern *within* a ward is real (§5A, "What those figures do NOT cover"). The second answer is largely no, it is disclosed in the product, and it matters more than the first for anyone reading the map block by block. Section 8 lists every source with a resolvable link.

Writing it prompted a proper review of the constants, which was overdue. That turned up **five** that were uncited, self-referential, or not scientifically supportable — all now corrected and sourced (§4). A validation harness scores model output against published measurements on every run, so the model can't drift quietly again. It passes **10 of 11** checks. The one failure is deliberate and live: §4.2.2 found that the evapotranspiration humidity ramp breaks the model's own physical bar on six of our 298 observed humidity readings. Capping the ramp cut that to three and left 230 of the 298 readings bit-for-bit unchanged; the harness reports the residual rather than hiding it.

What changed since the first issue. The previous version said plainly that nothing had been validated against measurement for our own wards. That is no longer true. We built a calibration set of **49 usable NASA ECOSTRESS scenes** over Kolkata (2024-01 → 2026-07, 179 acquisitions attempted, every exclusion recorded), attached hour-matched meteorology to each, and compared the model to observation scene by scene. Section 5A reports what that found, including the parts that did not go our way.

The headline result is a number we can now put on the tool:

	measured accuracy	status in the interface
Night	± 3.0 °C (50 ward-scenes)	labelled <i>calibrated</i>
Day	± 4.5 °C (29 ward-scenes)	labelled <i>indicative only</i>

Both figures improved in the July revision, and the night one changed in kind rather than degree. The model used to put the night surface *below* air temperature; measured, it sits 2.10 °C above — the nocturnal heat island is the defining signature of the thing we are selling, and we had its sign backwards. §5B has the diagnosis and the fix.

Daytime is worse for a reason that no amount of engineering will fix, and §5A explains it: the limit is the resolution of the weather data driving the model, not the physics inside it. Both figures are now shown next to the temperature readout on the tool itself, rather than living only in this document.

Two earlier findings are worth reading before the detail. The headline heat figure looked roughly eight times too high against published Kolkata values and was initially recorded as our most serious defect; it turned out to be **mislabelled rather than miscalibrated** (§4.1). And the attempt to fit the model's remaining free constants to measurement **failed its acceptance test** — reported in §5A rather than quietly retried, because the failure is informative.

Section 6 sets out what I still don't trust, without hedging.

1. What the Green Score computes

A 0–100 number, the equal-weighted mean of three sub-scores:

$$\text{Green Score} = 100 \times [\text{greening ratio} + \text{cooling achieved} + \text{budget efficiency}] \div 3$$

Each sub-score is capped at 1.0, so the total cannot be bought with spending. **All three are displayed raw in the tool** alongside the total, so the composite is auditable — you can see which component produced the number rather than taking it on trust. Berlin's BAF and the Seattle Green Factor both publish their components for the same reason.

Greening ratio

A **weighted-area ratio** — the same structure used by every established green-space planning standard: Berlin's **Biotope Area Factor** (mandatory since 1994), the **Seattle Green Factor**, and Singapore's **Green Plot Ratio**. All compute *sum of (area × ecological weight) ÷ total area*.

Surface	Weight	Basis
In-ground vegetation	1.0	Berlin BAF "vegetation connected to soil"
Tree canopy over street	0.6	Berlin / Seattle canopy bands
Green facade	0.6	Berlin wall greening 0.5; Seattle up to 0.7
Water / wetland	0.8	Berlin water surface
Cool roof	0.1	Our own low weight — a painted roof is not habitat
Sealed surface	0.0	All three standards

Normalised against **0.45**, the midpoint of Berlin's 0.30–0.60 target band.

Cooling achieved

Modelled surface-temperature drop versus the no-intervention baseline, normalised against 2.5 °C.

Budget efficiency

Degrees cooled per ₹ crore, normalised against 0.15 °C/crore.

2. Where every number comes from

Measured = from data · **Cited** = traceable to a named source · **Tool-relative** = normalised against our own output · **Placeholder** = explicitly temporary.

Costs

Item	Value	Grade	Source
Cool roof	₹150/m²	Cited — high	Telangana Cool Roof Policy 2023-28; NRDC India
Street tree	₹1,500/tree	Cited — med-high	Municipal planting + 3-yr maintenance
Pocket park	₹1.5 crore/ha	Cited — medium	Gujarat AMRUT 2.0 (2 named projects)
Green facade	₹9,500/m²	Cited — medium	Indian vendor pricing; excludes irrigation opex
Wetland	<i>no figure exists</i>	Open	See §6

Physical coefficients

Item	Value	Grade	Source
Dark roof albedo	0.15	Cited — high	LBNL Heat Island Group
Aged cool-roof albedo	0.60	Cited — high	LBNL <i>aged</i> value, not fresh-white 0.85 — deliberately conservative
Pocket park radius	50 m (0.785 ha)	Cited — Kolkata-specific	Li et al. 2022, threshold value of efficiency (TVoE 0.77 ha — genuinely Kolkata's; see §4.2 correction)
Evapotranspiration L	0.46 (shipped)	Re-derived §4.2.1 — an admissible choice, not a determined one; its humidity ramp has a defect, §4.2.2	Chosen inside a one-sided feasible region; the old two-constraint band is a fossil
Facade heat reduction	0.03	Cited — corrected from 0.30	Gunawardena & Steemers 2023
Influence kernel λ	≈ 47 m	Empirical, honestly labelled	See §4.4
Sky temperature	computed, not fixed	Cited	Brutsaert (1975) clear-sky emissivity, cloud-screened; replaced two hard-coded values
Night evapotranspiration	10 % of daytime	Cited	Stomata close after dark; tapers to zero below the dewpoint
Night anthropogenic heat	50 % of daytime	Estimated, bounded	Diurnal load profiles, South and East Asian cities
Warming pathways	+1.25 / +4.1 °C	Cited — corrected	Dhara et al. 2025, PLOS Climate

Score parameters

Item	Value	Grade
Weights	equal thirds	Corrected from 40/40/20 — see §4.5
Greening reference	0.45	Cited, but transplanted (see §6)
Cooling reference	2.5 °C	Cited-ish
Efficiency reference	0.15 °C/cr	Tool-relative — labelled, not fixed

Input geometry — genuinely measured

Microsoft ML Building Footprints (ODbL) · Google Open Buildings 2.5D heights (CC BY 4.0) · OpenStreetMap road centrelines (ODbL) · Norwegian Meteorological Institute live air temperature

Added for validation (§5A): NASA ECOSTRESS 70 m land-surface temperature (public domain) · NASA POWER hourly meteorology (public domain) · ESA WorldCover 10 m land cover (CC BY 4.0) · GHS-SMOD R2023A settlement classification (CC BY 4.0). All four are free and redistributable; none carries a non-commercial clause. Open-Meteo was the easier route for historical weather and was **rejected on licence grounds**, being non-commercial.

3. What the model produces

Scenario	Cooling	Cost	Score
Nothing	0.00 °C	₹0	2
Cool roofs 100%	2.25 °C	₹7.5 cr	69
Tree corridors max	2.01 °C	₹66 lakh	84
Green facades max	0.14 °C	₹190 cr	4
Everything	4.45 °C	₹203 cr	66

Trees score highest, and facades now score almost nothing. Both are consequences of the corrections, not of design. Cool roofs cost **₹3.3 crore per °C**; facades cost **₹1,382 crore per °C**. That ordering matches Indian Heat Action Plan practice, where cool roofs lead — and it emerges from cited costs rather than being put there.

Doing everything scores lower than trees alone (66 vs 84). Deliberate: the cooling term caps at 2.5 °C, so extra cooling earns nothing while cost destroys the efficiency term. It rewards cost-effectiveness. It is counterintuitive enough that the interface should explain it, and currently doesn't.

4. The audit — five corrections

4.1 The heat figure was mislabelled, not miscalibrated

The tool displayed "**UHI Δ vs rural: +11 °C**". Published surface UHI for Kolkata is **0.85–1.5 °C** (Nayak 2023, Jain 2023, Siddiqui 2021). That looks like an eight-fold overstatement, and I initially wrote it up as our most serious defect.

It isn't. Voogt & Oke (2003), the canonical reference, put clear-sky **daytime surface** UHI at **10–15 °C**, while canopy-layer **air** UHI is 2–5 °C. The published Kolkata figures are 1 km MODIS, season-averaged, measured against a real rural population. Ours is clear-sky midday at 7 m against a synthetic all-vegetation cell. **Three different quantities.**

The number is defensible. The label was not: "UHI" and "rural" are both load-bearing terms we had no claim to.

Fixed: renamed to "**Δ vs all-green ref**", with the reference stated in the tooltip and an explicit note that measured Kolkata SUHII is a different quantity.

2026-08-07 implementation correction: the shared `equilibriumC()` reference now includes night storage (`store`). The duplicated old formulas omitted it, overstating retained/night contrast but not the heat field. Explore and Compare now share `greenReferenceContrastC()` and record `heat-metrics-v2` ; this is not a recalibration of `heat-model-v1` or the observed rural validation baseline.

I'm flagging this prominently because it is the error I would most want caught in my own work: I compared two numbers that shared a name and not a definition.

4.2 Evapotranspiration was over-strong — now derived, not tuned

The ET coefficient was 0.50, which put a fully-vegetated surface **5.8 K below air temperature** — more cooling than evapotranspiration can physically deliver.

Rather than pick a nicer number, I solved for the value satisfying **two independent published constraints at once**:

- park cool-island intensity of **4.83–8.07 °C** (Mitra et al. 2022, measured in Kolkata)
- vegetated surface no more than **4 K below air** (physical ceiling on ET)

They intersect at **0.40–0.46**. This derivation picked **0.43**, the midpoint. Both constraints passed. *The shipped constant is 0.46, not 0.43 — see the correction below. And the band itself does not survive: §4.2.1 shows it cannot be reproduced from the current model at all.*

CORRECTION 2026-08-08 — the first constraint above is misread, and §4.1's own warning is what it violates.

Two errors, found when the citation was checked against the source:

1. **The paper is not by Mitra.** DOI `10.3389/fenvs.2022.1073914` is Li, Lu, Fu, Sun, Pan, Han, Guo & Li (2022), *Diverse cooling effects of green space on urban heat island in tropical megacities*. There is no Mitra on the author list.
2. **4.83–8.07 °C is not a Kolkata band.** **8.07 °C is Kolkata's maximum UCI**; **4.83 °C is Bangkok's maximum**. It is a cross-city range of daytime, winter maxima from fused Landsat/MODIS — not a range of park cooling measured within Kolkata.

What from that paper is genuinely Kolkata's, and stands: **TVoE 0.77 ha** and **reach 420 m**.

This is precisely the error §4.1 flags as "the one I would most want caught in my own work" — two numbers that share a name and not a definition. There it was a validation baseline; here it is a cross-city range of *maxima* compared against a *within-ward* cooling value. Recording it in the same terms rather than quietly repairing the citation.

What it does to L . The real constraint from that paper is one-sided — Kolkata's daytime maximum is $8.07\text{ }^{\circ}\text{C}$ — so the model's $5.80\text{ }^{\circ}\text{C}$ local park cooling does not *violate* it. But the **lower bound was never a constraint**, and it was half of what pinned the intersection. L therefore rests on one valid constraint ($\leq 4\text{ K}$ below air) plus one that does not exist as stated, and **is no longer uniquely determined by the evidence as written**.

And the published L was not the shipped L . The derivation above produced **0.43**, the midpoint of `[0.40, 0.46]`, and this document stated 0.43 in two places — including in the first draft of this very correction. The constant in `src/scripts/climate-engine/types.ts` is `L: 0.46`, moved later to the *top* of that band. Both numbers are now recorded here: **0.46 is what ships and what every figure in §5 is computed from**; 0.43 is what §4.2's derivation concluded. The two disagree, and *both* depend on the band whose lower bound is withdrawn — "the midpoint of `[0.40, 0.46]`" and "the top of `[0.40, 0.46]`" are equally unsupported once $4.83\text{ }^{\circ}\text{C}$ is gone. The constants table now states the shipped value.

L has NOT been changed. Re-deriving it means re-running the calibration and re-validating everything downstream — a code change, not a documentation one, and one that must not be made casually: **cooling is $\frac{1}{3}$ of the Green Score**, so if L moves, the scores move with it. The honest state today is a shipped constant with a partly-withdrawn derivation, recorded here rather than hidden.

4.2.1 The re-derivation, 2026-08-09 — the value stands, the reasoning did not

The correction above left L in an unsatisfactory state: a shipped constant whose stated derivation was half-withdrawn. This is the follow-up it asked for. Three findings, and the third is worse than the original error.

1 · Nothing fits L . `fit-physics.py` fits `Q_day`, the `kRad:h` ratio and Brutsaert `c` — L is not among them. In `fit-ward-scale.py`, `l_et` is a *bounded* parameter, and `data/calibration/ward-scale-fit.json` records it as pinned in **every** candidate. It lands on 0.6, 0.8, or 0.46 — always the ceiling it was handed. The unconstrained fit wants about **0.8**, roughly twice what evapotranspiration can deliver.

2 · So the withdrawn lower bound never bound anything. A parameter that rails upward is decided by its ceiling alone. Losing the $4.83\text{ }^{\circ}\text{C}$ figure destroys the *rationale* — "the midpoint of the band", later "the top of the band" — without making 0.46 inadmissible. The value and the argument for it failed independently, and only the argument was ever wrong.

3 · And `[0.40, 0.46]` cannot be reproduced from this model at all. Local park cooling in the shipped model is exactly `15 · L_eff`. Feed it even the *old*, two-sided `4.83–8.07` $^{\circ}\text{C}$ band and it maps to $L \in [0.383, 0.640]$ at $\text{rh } 60$ — **not `[0.40, 0.46]`**. That band is a fossil of an earlier model version, predating the Q retune and the humidity scaling of L . It was never this model's constraint, and should not be cited as one by anybody, us included.

What actually constrains L . Two ceilings, both one-sided: park cooling $\leq 8.07\text{ }^{\circ}\text{C}$ (Li et al. 2022, Kolkata daytime maximum) and a vegetated surface within 4 K of air. Translating them into constant-space depends on humidity, because `currentParams` scales L by `evapScale(rh)` (see §4.2.2) — which moves it substantially across the range:

requirement holds at	binding bar	ceiling on the constant L
rh 60 % (the fallback, and the only point the harness used to check)	park cooling	$L \leq 0.640$
rh 30 %	vegetated surface	$L \leq 0.515$

Which bar binds is not a constant. Below roughly 35 % rh the vegetated-surface bar is the tighter of the two. The first draft of this section derived every ceiling from park cooling alone and put rh 30 at `0.527` ; that was wrong, and the correct figure is `0.515` .

4.2.2 A defect found while verifying §4.2.1 — the humidity ramp has no upper anchor

Checking the above properly turned up something the re-derivation had missed, and it is a model defect rather than a documentation one.

`currentParams` scales L by `evap = 0.6 + 0.6 · (1 - rh/100)` . That keeps **raising** evapotranspiration as the air dries, without limit, reaching $1.2\times$ the constant as $rh \rightarrow 0$. Meanwhile the 4 K headroom **shrinks** as the sky dries, because a dry sky radiates less back and `tSky` falls. The two curves cross at about **22 % rh** .

This is reachable on observed weather. `data/calibration/ward-observations.json` records 298 humidity readings for these wards: median 54.3 %, 5th percentile 25.9 %, **minimum 14.1 %** . Six of those 298 readings put a vegetated surface further than 4 K below air — the model violating its own physical bar on weather we measured. Imposing the 48 °C heatwave overlay widens the window; it does not cause it.

The physics points the other way from the ramp: under high vapour-pressure deficit real plants **close their stomata and transpiration falls** . A linear increase with dryness is a reasonable local approximation near typical humidity and an extrapolation with no anchor at the dry end — which is precisely where it breaks.

The defect is the ramp, not L 's magnitude. The options are not equivalent and none is free:

option	readings changed (of 298)	violations left, no heatwave	violations left, any heatwave (of 1490)
today — uncapped ramp, L 0.46	—	6	134
cap <code>evap</code> at 1.00	68	3	15
cap <code>evap</code> at 0.95	100	0	9
lower L to 0.40, ramp uncapped	all	0	12

Capping is the more surgical instrument — it leaves every reading above the cap bit-for-bit unchanged, so median-case Green Scores do not move at all — but a cap tight enough to clear the worst corner (dry base plus extreme heatwave) bites below 44 % rh and touches a third of the archive. Lowering L moves every score instead.

Decision: cap the ramp at 1.0. It fixes the physics where it is wrong and leaves it alone where it is right — the cap binds only below rh 33.3 %, so **230 of the 298 readings are bit-for-bit unchanged** and the median-case Green Score does not move at all. Lowering `L` would instead move every score on every day, including the median case that was never wrong, and would treat a magnitude as the fault when the shape is.

Measured outcome:

	before	after
crossing (bar breaks below)	22 % rh	16 % rh
violations on observed weather	6 of 298	3 of 298
violations including the heatwave overlay	134 of 1490	15 of 1490
readings changed at all	—	68 of 298

Not fully closed, deliberately. Below the cap the ET term is frozen while the 4 K headroom keeps shrinking with the drying sky, so a crossing must still exist somewhere. Closing it entirely needs a cap near 0.95, which bites below 44 % rh and reshapes a third of the archive — humidity that is entirely ordinary for Kolkata. Three residual readings out of 298 is the better trade, and `validate-model.mjs` **fails visibly** on them rather than hiding them.

One consequence worth stating plainly: this moved two shipped constants. The ramp is not only in the renderer — `scripts/fit-ward-scale.py` trains on the same expression, and the shipped `Q` and `STORE_NIGHT` are candidate G's fitted values from that script. A model calibrated against one shape and executed with another is not calibrated, so the fit was re-run under the cap and its output adopted:

constant	was	now
<code>Q</code> (anthropogenic heat, day)	0.4131	0.3936
<code>STORE_NIGHT</code> (nocturnal release)	0.1052	0.1081

Adopting them costs **zero** movement in the Green Score on modest, moderate and full intervention. Built-surface-above-air goes 13.34 → 13.02 K and land-cover contrast 9.62 → 9.46 °C, both well inside their published bars. `data/calibration/model-accuracy.json` was re-measured against the new constants; every stratum moved by less than 0.13 K and the published bands (± 3.0 night, ± 4.5 day) remain conservative against the measured error.

The expression now has one definition per language. It had been copy-pasted across seven sites — two in the renderer, five in `scripts/` — and capping only some would have left the offline analyses validating physics the page does not run. It is now `evapScale()` in `heat-map-model.ts` and `evap_scale()` in `scripts/_physics.py`, with every site calling one of those two.

Why it was not moved to the ceiling. Because a ceiling is not evidence that the value belongs against it. Raising `L` to 0.527 would mean adopting the largest number the guardrail permits on the say-so of a fit that wants 0.8 — and `fit-ward-scale.py`'s own verdict on that temptation is *"the same mistake in a nicer suit"*.

The cost of being wrong here is smaller than §4.2 implied. That correction warned that if `L` moves, every score moves with it. Directionally true, but now measured: taking `L` from 0.46 to 0.527 shifts the Green Score by **+1 point on a modest scenario, +2 on a moderate one, and exactly 0 at full intervention** — out of 100. The published scores are far less sensitive to this constant than the earlier warning suggested, and saying so is part of the correction.

What changed in the code. Only the validation harness, and only to make it honest: both ET bars were single-point checks at rh 60 on a coefficient that moves substantially with humidity. `validate-model.mjs` now sweeps the reachable space — every heatwave the instrument can produce, plus the humidity at which each bar first breaks — and reports that break point as a robustness margin rather than asserting a Kolkata humidity floor nobody measured.

Standing conclusion. `L = 0.46` is an *admissible choice inside a one-sided feasible region* for the humidity range that carries the great majority of the weather, not a uniquely determined constant — and §4.2.2 records where that region runs out. It is stated that way in `src/scripts/climate-engine/types.ts` and here, and it should not be described with more precision than that anywhere else.

4.3 Green facades were overstated roughly tenfold

The facade heat-reduction coefficient was **0.30** with no source. The only neighbourhood-scale measurement I could find — Gunawardena & Steemers 2023, *Buildings & Cities* — reports green facades moving heat-island intensity **1.86 K → 1.81 K, about 3%**, and cutting space-conditioning energy 2.1%. The dramatic -13 to -20 °C figures in circulation are **local wall-surface** effects; the same authors found the vapour flux "advects away to background levels".

Fixed: coefficient **0.30 → 0.03**, cited. I also **deleted** the facade ground-vegetation term entirely — its factor of 0.15 had no measurement support, and crediting a wall treatment with ground vegetation was double-counting.

Consequence: facades went from an apparently competitive lever (1.90 °C) to a marginal one (0.14 °C). That is what the literature says they are.

4.4 A physical constant was tuned for appearance — now honestly named

The diffusion constant controlling how far cooling spreads was selected because it produced a **legible map**, not because a measurement supported it. Our own documentation recorded this openly, which I credit, but a visual criterion had chosen a physical parameter.

It turns out the deeper problem is that it was never diffusivity at all: **lateral heat conduction between 7.3 m cells is roughly eight orders of magnitude too small to matter** (soil diurnal damping depth ≈ 0.12 m). Real horizontal coupling comes from wind-driven advection, which this model does not represent.

Fixed: relabelled throughout as an **empirical spatial-influence kernel**, with the physics stated plainly. The value is unchanged — deliberately short, because our field is surface temperature, which real thermal imagery shows as sharp. The 120–300 m park-cooling distances in the literature are an air-temperature phenomenon.

4.5 Two pathway figures were scientifically indefensible

The projection slider offered $-1.2\text{ }^{\circ}\text{C}$ ("target 2030") and $+2.4\text{ }^{\circ}\text{C}$ ("BAU 2040"), both uncited.

No emissions scenario produces regional cooling over India. Even the most aggressive mitigation pathway has India warming through mid-century because of committed warming. A "target" showing Kolkata $1.2\text{ }^{\circ}\text{C}$ cooler than today is a mitigation aspiration drawn on a physical-temperature axis, and it reads as a forecast.

Fixed: the negative pathway is **deleted**. The remaining two are cited to Dhara et al. 2025, *PLOS Climate* (post-AR6 India update): **SSP2-4.5 $+1.25\text{ }^{\circ}\text{C}$ (2041–2060)** and **SSP5-8.5 $+4.1\text{ }^{\circ}\text{C}$** .

Also corrected: the score weights

The 40/40/20 split had **no published precedent** and implied greening and cooling were each exactly twice as important as cost. Equal weighting is the most commonly applied approach in composite-indicator practice (OECD/JRC 2008) and the convention in urban-resilience indices. **Now equal thirds**, cited. An arbitrary split dressed as a derived one is worse than a plain one.

5. The validation harness

`scripts/validate-model.mjs` scores model output against cited benchmarks on every run.

Values below are the harness's own output, re-read on **2026-08-08** (the table had drifted six rows stale against the shipped constants).

Check	Value	Expected	
Land-cover thermal contrast	$9.46\text{ }^{\circ}\text{C}$	$8\text{--}18\text{ }^{\circ}\text{C}$	ok
Max ward cooling $\leq$ local park cooling	$4.54\text{ }^{\circ}\text{C}$	$\leq 5.80\text{ }^{\circ}\text{C}$	ok
Local park cooling	$5.80\text{ }^{\circ}\text{C}$	$\leq 8.07\text{ }^{\circ}\text{C}$ (Kolkata daytime max)	ok — one-sided, see §4.2
Vegetated surface below air	0.48 K	$\leq 4\text{ K}$	ok
Built surface above air	13.02 K	$5\text{--}25\text{ K}$	ok
Facade vs cool-roof cost per m^2	$63\times$	$40\text{--}80\times$	ok
Score at zero intervention	2	$0\text{--}10$	ok
Score at full intervention	67	$0\text{--}100$	ok
Park-cooling bar on observed weather	0 readings	0	ok
Heatwave overlay from the median day	2.54 K	$0\text{--}4\text{ K}$	ok
Vegetated-surface bar on observed weather	3 readings	0	FAILS — §4.2.2

10 of 11, from 4 of 10 before the audit. The single failure is deliberate and live: the three rows in bold were added in the 2026-08-09 re-derivation, and the last of them is the defect §4.2.2 describes — the evapotranspiration humidity ramp breaks the model's own 4 K physical bar in dry air. Capping the ramp took it from six of 298 observed readings to three; the rest cannot be closed without reshaping ordinary Kolkata humidity, so it is reported rather than relaxed, which is what this harness is for.

One methodological note you should interrogate. Three checks were demoted from pass/fail to informational. Before doing that I tested whether they were real defects, and all three borrowed **air-temperature** benchmarks for a model computing **surface** temperature. The clearest evidence: WRI India's figures for the *same* street trees are **−0.3 °C air** and **−27.5 °C road surface** — a 90× gap. No published ward-mean *surface* benchmark exists, so max cooling is now bounded by a physically necessary rule instead (a field's mean cannot shift further than its largest local change). "My benchmark was wrong" is what motivated reasoning sounds like, so the reasoning is written into the code for you to check.

5A. Validation against measurement — what it found

This section is new. Everything above it checks the model against *published literature*; this checks it against *satellite observations of our own three wards*.

The calibration set

Source	NASA ECOSTRESS L2T LSTE v002, 70 m land-surface temperature
Window	2024-01-01 → 2026-07-01, both day and night overpasses
Attempted	179 acquisitions
Usable	49 scenes (31 night), after cloud and coverage screening
Excluded	130, each recorded with its reason — no silent exclusions
Urban / rural classes	GHS-SMOD R2023A (EU/UN Degree of Urbanisation), not a temperature threshold
Meteorology	NASA POWER hourly, matched to each overpass in local solar time
Land cover	ESA WorldCover 10 m, measured inside the same masks — not assumed

Two controls were run rather than assumed. The urban–rural **view-angle** difference correlated with the measured heat-island signal at $r = -0.322$ ($p = 0.024$), meaning about 10 % of the apparent signal was sensor geometry rather than climate; restricting to near-nadir scenes ($\leq 0.75^\circ$) removes it and retains 36 scenes. The three alternative definitions of "rural" disagree by only 0.08 °C, so that choice is not a material source of uncertainty.

How accurate the model is

The important question is not only "how close is our model" but "**how close could any model get** with the data available". So alongside the model's own error we computed a **ceiling**: the error of the best possible statistical predictor built from the same weather inputs, scored leave-one-out so it cannot flatter itself.

	ward-scenes	best achievable	our model	reported
Night	50	2.23 °C	2.93 °C	± 3.0 °C
Day	29	3.34 °C	4.42 °C	± 4.5 °C

Revised July 2026. These are measured at WARD scale — see §5B, which explains why the previous mask-scale figures described a different surface than the one the tool renders.

Read the "best achievable" column first. **At night the model has real headroom** — a better model could reach ~2.2 °C, so continued work is worthwhile. **By day, no model driven by this weather data can beat ~3.3 °C.** Daytime surface temperature depends on cloud timing, local sunshine and soil moisture that a 50 km reanalysis grid cell simply cannot see. That is a property of the input data, not a defect we can engineer away, and the only real fix is higher-resolution meteorology.

This is why the tool now labels the daytime view **indicative only** and reserves quantitative language for night. Presenting a ± 5 °C daytime figure as decision-grade would be the actual inaccuracy.

What those figures do NOT cover — spatial skill

Everything in the table above is a **ward-mean** error: how far the model's average temperature sits from ECOSTRESS's average over the same 1400 m box. It says nothing about whether the pattern *inside* the ward is right — and the pattern is what a reader looking at a map actually uses. A model that got the ward average exactly right while placing every hot block in the wrong street would score identically in that table.

So we measured the other thing, at ECOSTRESS's native 70 m, with the ward mean removed from both sides so this is pattern only. 81 ward-scenes across 32 near-nadir scenes and three wards:

predictor	overall r	day	night
Our model	0.113	0.175	0.074
Built fraction alone	0.037	-0.012	0.067
Vegetation alone	0.229	0.297	0.186

A plain vegetation map predicts within-ward heat about twice as well as our full physics model. The two null predictors are the reason that is legible at all: $r = 0.11$ on its own reads as "some skill", and only the comparison shows it is worse than a single one of the layers the model is built from. Publishing the correlation without the nulls would have been technically true and materially misleading.

Decomposing the field into its three spatially-varying terms says why:

term	correlation with measured heat	spatial variation it contributes
Solar / albedo	0.040	0.02 °C
Built fraction	0.037	1.46 °C
Vegetation	0.229	0.53 °C

The built-fraction term contributes most of the picture and carries no measurable skill; the vegetation term carries the skill and is outweighed roughly three to one. So the map's block-by-block detail is largely a rendering of building-footprint density, and footprint density does not predict where ECOSTRESS sees heat at 70 m. The likely reason is physical rather than a coding error: in a dense ward nearly every cell is built, and daytime surface temperature is set by roof material and shading, which a footprint polygon does not carry.

This is the second independent line of evidence pointing at the model's structure rather than its constants. The fit above ran all three free parameters to their bounds; this shows the term that dominates the spatial field has no skill. Both point the same way, and together they say that tuning constants is not where the remaining error is.

What this does not affect. Ward-level accuracy is measured separately and is unchanged, as are the DC-URS resilience scores — the score reads ward-level indicators and never the per-cell field. The finding bears on one claim only: that the hot and cool blocks *within* a ward mean something.

What we did about it. The tool now states it. The colour legend carries the measured figure, the honesty banner says "block detail illustrative" at every screen width, and both readout tooltips carry the full sentence including this vegetation comparison. The figures are held as data in `src/scripts/climate-engine/accuracy.ts` and asserted, so the wording cannot go on claiming the pattern is unvalidated after that stops being true.

Method and its limits, for the record: near-nadir scenes only; cells dropped where ECOSTRESS is cloudy, low-quality or water; the model evaluated at equilibrium with no lateral diffusion, so real advection between adjacent cells counts against it; a ward is about 21×21 cells at 70 m, so a single scene's correlation is noisy and the aggregate is the figure. Reproduce with `python3 scripts/measure-spatial-accuracy.py`; full per-scene output in `data/calibration/spatial-accuracy.json`.

The fit failed, and that is the honest result

We then attempted to fit the model's three remaining free constants — anthropogenic heat, the radiative-to-convective coupling ratio, and the sky-emissivity coefficient — to the measured scenes, with residuals taken on urban *and* rural absolute temperatures rather than on their difference alone. Matching a difference while both sides are wrong is not a calibration.

It did not pass. **All three constants ran to the edge of their physically defensible bounds** and the error stayed at 3.75 °C against a ± 2 °C acceptance bar. Every one moved in the direction that makes the modelled surface warmer, and it was still too cold. Widening the bounds would have produced a number rather than a calibration, so the fitted values were **not adopted** — the model ships with its previous, literature-derived constants, and the fit output is recorded with an explicit "do not ship" flag.

Three candidate explanations were tested and none survived:

Hypothesis	Result
Missing heat-storage / thermal inertia	Error moved only 3.75 → 3.41 °C. Insufficient.
Evapotranspiration limited by soil moisture rather than air dryness	Removed the bias but <i>worsened</i> the scatter; the parameter ran to its bound, so it merely reduced evaporation overall.
Vegetation vigour (NDVI) explaining the daytime signal	Ruled out earlier — rural vegetation index exceeds urban in every scene regardless of the sign of the measured heat island.

What the evidence does point to is that **the evapotranspiration coefficient is too strong** — the one set in §4.2 from local park-cooling measurements. Regional satellite means disagree with local park measurements, which are genuinely different quantities (a park interior versus a mask-wide average). That tension is unresolved and is listed in §6 rather than split down the middle.

Why the acceptance bar itself was wrong

The ± 2 °C bar was written before anyone knew what was achievable. The ceiling analysis shows it was never reachable for daytime and sits at the edge of possible for night. A target that no model can meet is not a quality standard; the per-phase figures above replace it.

5B. Why the fit failed, and what fixed it (July 2026)

§5A above records a calibration that ran all three of its free constants to the edge of their bounds and stopped, with the verdict "*the model structure is wrong, not its constants.*" That was half right. The structure did need two changes — but the reason nothing worked was that **the fit's target was the wrong geography**.

The masks were not what they were called

The old fit scored the model against two GHS-SMOD classes: an "urban" mask of 3,363 km² and a "rural" one of 1,568. We sampled both with Sentinel-2, sixteen windows each, and they are the same landscape:

mask	area	vegetation (FVC)	built
"urban"	3,363 km²	0.678	0.16
"rural"	1,568 km²	0.654	0.01
the wards we actually render	1.4 km each	0.31 – 0.45	0.22 – 0.37

The "urban" side is marginally the *greener*. Their measured temperature difference is 0.34 °C, and now we know why: it is the difference between delta with villages and delta with crops, not an urban heat island. Meanwhile the wards are twice as built and half as vegetated as the mask named "urban". We were fitting one

question and shipping the answer to another, and no constant can make a two-thirds-vegetated landscape behave like a city block. That is why every parameter ran to its bound.

The calibration set is now built at ward scale: 79 ward-scenes of ECOSTRESS at its native 70 m inside each ward footprint, paired with that ward's own measured vegetation, albedo and building coverage — the same three numbers the browser draws with.

Two structural faults, both real

The night sign was wrong. Measured, the ward surface sits **2.10 °C above** air at night — stored daytime heat discharging, which is the nocturnal heat island. The model put it 3.45 °C *below*, because a steady-state balance computes a weighted mean of air and a sky 10–20 °C colder with nothing holding the surface up. A sign error is not a tuning problem. It needed the storage term (ΔQ_s) that a steady-state formulation omits by construction.

No lever could warm a vegetated surface. The evapotranspiration coefficient was held fixed at a value derived from park-*interior* cooling measurements and applied to an area mean — the tension §6 already flagged as unresolved. Freed inside its defensible range, it moved to the top of that range.

The result

	old structure	shipped	best achievable
Night	3.27 °C (bias −1.54)	2.93 °C (bias +0.18)	2.23 °C
Day	4.91 °C (bias +2.55)	4.42 °C (bias +0.91)	3.34 °C

Out-of-sample, holding one ward back and fitting on the other two: 3.95 → 3.62 °C.

Night's bias is the headline — it moved from the wrong side of air temperature to the right one. Day improved but sits 1.08 °C above its ceiling, so the daytime structure is still incomplete and that gap is ours, not the data's.

What we refused

The unconstrained fit wanted a sky-emissivity coefficient of 1.40 — the top of the published range — and an ET coefficient roughly twice what evapotranspiration can physically deliver. Both were refused. Our own invariants caught them: at Kolkata humidity a coefficient of 1.40 makes the clear sky as warm as the air, and a second check requires the modelled night sky at 28 °C / 80 % to land 19–21 °C, which holds only in a narrow window around the published 1.24.

Holding every constant inside a defensible range simultaneously costs about 0.13 °C of accuracy against the unconstrained fit. We paid it. A model that fits better using a constant nobody can defend is not a better model, and a client is entitled to ask where each number came from.

The within-ward pattern improved but is still not validated

Halving the anthropogenic-heat term cut the built-fraction term's spatial dominance from 1.46 °C to 0.82 °C, now comparable to vegetation's 0.62 °C rather than three times it. Spatial correlation rose from 0.113 to 0.162. It remains below the 0.229 of a plain vegetation map, so the disclosure on the tool stands unchanged. A ward-mean fit cannot fix this — building coverage is a single number per ward in that fit, so nothing in it constrains how the field varies *inside* a ward. That needs a per-cell calibration, which the cached observations now make possible.

6. What I still don't trust

- **The model is now validated against measurement, and it did not pass cleanly.** This replaces the previous version's statement that nothing had been checked against our own wards. It has been (§5A, §5B), and the result is ± 3.0 °C at night and ± 4.5 °C by day — usable for screening and ranking interventions, not for anything requiring a defensible absolute temperature. The daytime figure cannot be improved without better weather data.
- **The map's within-ward pattern is not validated, and is weaker than a vegetation map.** Measured against ECOSTRESS at 70 m (§5A), our field scores $r = 0.11$ where vegetation cover alone scores 0.23. The built-fraction term contributes 1.46 °C of spatial variation against vegetation's 0.53 °C while correlating at 0.037, so the block-level detail is mostly footprint density rendered as heat. It is disclosed in the product. Ward-level figures and the DC-URS scores are unaffected. Fixing it is a change to the model's structure, not its constants — which is the same conclusion the failed fit reached independently.
- **The evapotranspiration coefficient is contested by our own measurements.** §4.2 derives $L = 0.43$ from local park-cooling studies (the shipped constant is **0.46**, and half of that derivation is withdrawn — see the §4.2 correction). The satellite fit says it is too strong. These measure different things — a park interior versus a ward-wide average — so this is a real conflict between two valid datasets, not an error to be averaged away. It needs a deliberate decision.
- **The greening benchmark is transplanted.** Berlin's 0.30–0.60 band applies to *individual development sites in a temperate European city*. We apply it to *whole wards in tropical Kolkata*. Convenient, not validated.
- **Efficiency is still tool-relative.** No published cost-per-degree-cooled benchmark exists, so a ward can only score well relative to what our own model produces. Now labelled as such rather than presented as grounded.
- **No Indian wetland cost figure exists.** Government schemes fund *management*, not per-hectare creation. Also worth knowing: East Kolkata Wetlands sit under the 2006 Conservation Act, which prohibits conversion — so a "create wetland" lever may be modelling a legally impossible action. The lever is currently retired from the interface.

- **Several citations were verified from abstracts only.** Re-verification before any client deliverable has not happened.
 - **A licensing question for you:** our building footprints are ODbL, which carries share-alike obligations for derived *databases*. Rendered maps are almost certainly fine; the derived data files we ship may not be. Worth a proper answer before anything paid.
-

7. Open decision

Whether the Green Score should retain the **cooling term**, now that we know how uncertain it is.

The previous version of this document posed this question while the accuracy was unknown. It is now measured: ± 3.0 °C at night, ± 4.5 °C by day. That changes the question from "*can we trust it?*" to "*is this precision good enough for what we sell?*"

A score built only on the greening ratio and cost would be narrower but fully defensible today — both rest on published standards and cited Indian unit costs. Including modelled cooling makes the score more useful and less certain, and it is currently two of the three components (cooling directly, plus efficiency, which is cooling divided by cost).

Two things worth weighing. In its favour: the score uses cooling as a **relative** quantity — ranking interventions against each other under identical forcing — and much of the absolute error is common to both sides of that comparison, so the ranking is more reliable than the $\pm$ figure suggests. Against it: we cannot currently prove that, and a client is entitled to ask.

A middle option exists that was not available before: retain the cooling term for the **night** phase, where the model is calibrated, and treat the daytime contribution as indicative — matching what the interface already tells users.

This is a judgement about what the firm is willing to stand behind rather than a technical question, so it sits with the partners rather than with engineering.

8. Sources

Verification status is marked per item: **[V]** full text or primary document consulted · **[A]** abstract or catalogue record only, not yet fully verified.

Scoring standards and index methodology

Source	Used for	Link
Berlin Biotope Area Factor (Senate Dept. for Urban Development) [V]	Greening weights; 0.30–0.60 target band	berlin.de/sen/uvk/en/nature-and-green/landscape-planning/baf-biotope-area-factor
Seattle Green Factor (SDCI) [V]	Cross-check on surface weights	seattle.gov/sdc/codes/codes-we-enforce-(a-z)/seattle-green-factor
Singapore Green Plot Ratio (URA LUSH) [A]	Third comparator on the weighted-area form	ura.gov.sg/Corporate/Guidelines/Development-Control/Non-Residential/EI/Greenery
OECD/JRC (2008), <i>Handbook on Constructing Composite Indicators</i> [V]	Equal weighting; sensitivity-analysis requirement	doi:10.1787/9789264043466-en

Physics, cooling evidence, and the surface-vs-air distinction

Source	Used for	Link
Voogt & Oke (2003), <i>Thermal remote sensing of urban climates</i> , <i>RSE</i> 86(3):370–384 [V]	§4.1 — daytime <i>surface</i> UHI 10–15 °C vs canopy <i>air</i> UHI 2–5 °C	doi:10.1016/S0034-4257(03)00079-8
Li, Lu, Fu, Sun, Pan, Han, Guo & Li (2022) , <i>Diverse cooling effects of green space on urban heat island in tropical megacities</i> , <i>Frontiers in Environmental Science</i> [V] — previously miscited here as "Mitra et al."	§4.2 — TVoE 0.77 ha and reach 420 m are Kolkata's and stand. Kolkata's daytime maximum UCI is 8.07 °C; 4.83 °C is Bangkok's — the "4.83–8.07 °C Kolkata band" this register formerly claimed does not exist. See the §4.2 correction.	doi:10.3389/fenvs.2022.1073914
Gunawardena & Steemers (2023), <i>Neighbourhood-scale vertical greening</i> , <i>Buildings & Cities</i> 4(1) [V]	§4.3 — heat-island intensity 1.86→1.81 K (~3%); energy 2.1–5.2%	doi:10.5334/bc.282
LBNL Heat Island Group — cool roof materials [V]	Albedo 0.15 dark / 0.60 aged-cool	heatisland.lbl.gov/cool-science/cool-roofs
WRI India , <i>Urban Trees' Cooling Potential</i> [V]	Canopy dose-response; the –0.3 °C air vs –27.5 °C surface contrast in §5	wri.org/insights/urban-trees-cooling-potential

Climate projections

Source	Used for	Link
Dhara, Deshpande, Roxy, Dalpadado & Shrestha (2025), PLOS Climate 4(11):e0000724 [V]	§4.5 — SSP2-4.5 +1.25 °C (2041–60); SSP5-8.5 +4.1 °C	doi:10.1371/journal.pclm.000724
Ministry of Earth Sciences (2020), <i>Assessment of Climate Change over the Indian Region</i> [A]	Cross-check: RCP4.5 +2.41 ± 0.40 °C by 2066–95	link.springer.com/book/10.1007/978-981-15-4327-2

Urban heat island benchmarks (used to sanity-check, not to calibrate)

Source	Kolkata / India value	Link
Nayak, Vinod & Prasad (2023), Applied Sciences 13(24):13323 [A]	Kolkata night SUHII 0.85 °C annual	doi:10.3390/app132413323
Jain (2023), Frontiers in Sustainable Cities 5:1084573 [A]	Kolkata night 1.3–1.5 °C (DJF)	doi:10.3389/frsc.2023.1084573
Siddiqui et al. (2021), Sustainable Cities and Society 75:103374 [A]	Indian metros night 1.34–2.07 °C	doi:10.1016/j.scs.2021.103374
Peng et al. (2012), Environmental Science & Technology 46(2):696–703 [A]	Global 1.5 °C day / 1.1 °C night; equal-area rural reference method	doi:10.1021/es2030438
Chakraborty & Lee (2019), Int. J. Applied Earth Obs. & Geoinformation 74:269–280 [A]	Global 0.85 day / 0.55 night; water-masking rationale	doi:10.1016/j.jag.2018.09.015
Shastri et al. (2017), Scientific Reports 7:40178 [V]	Indian daytime <i>cool</i> island in pre-monsoon — why Kolkata needs explaining	doi:10.1038/srep40178
Kumar et al. (2017), Scientific Reports 7:14054 [A]	60% of 89 Indian urban areas show a daytime cool island	doi:10.1038/s41598-017-14213-2

Cost sources (all Indian)

Source	Used for	Link
Telangana Cool Roof Policy 2023–2028 [V]	₹150/m ² cool roof; indoor –2.1 to –4.3 °C	telangana.gov.in (policy PDF)
NRDC, <i>Keeping It Cool: How India Can Protect Its People with Cool Roofs</i> [V]	Ahmedabad lime-wash rate; surface up to –30 °C	nrdc.org/resources/keeping-it-cool-how-india-can-protect-its-people-cool-roofs
Gujarat AMRUT 2.0 urban gardens (2026) [A]	₹1.5 crore/ha park capex, from two named projects	Press release; see <code>heat-map-intervention-model.md</code> §9
PreventionWeb / NRDC — Ahmedabad Heat Action Plan [A]	>1,100 deaths avoided/yr; –27% mortality on hottest days	preventionweb.net

Input data and licences

Dataset	Licence	Link
Microsoft ML Building Footprints	ODbL — share-alike, see §6	github.com/microsoft/GlobalMLBuildingFootprints
Google Open Buildings 2.5D	CC BY 4.0	sites.research.google/open-buildings
OpenStreetMap road centrelines	ODbL	openstreetmap.org/copyright
Norwegian Meteorological Institute (live air temp)	Free, keyless	api.met.no
NASA ECOSTRESS L2T LSTE v002	US public domain	doi:10.5067/ECOSTRESS/ECO_L2T_LSTE.002
ESA WorldCover 10 m v200	CC BY 4.0	doi:10.5281/zenodo.7254221
JRC GHS-SMOD R2023A (Degree of Urbanisation)	CC BY 4.0	doi:10.2905/A0DF7A6F-49DE-46EA-9BDE-563437A6E2BA
NASA POWER hourly meteorology (MERRA-2)	US public domain, no key	power.larc.nasa.gov
Open-Meteo historical archive	Non-commercial — rejected	open-meteo.com/en/license

Attribution strings required by licence are recorded in `docs/heat-map-intervention-model.md` and rendered on the tool itself.

Methods used in the validation (§5A)

Source	Used for	Link
Brutsaert (1975), <i>Water Resources Research</i> 11(5):742–744 [V]	Clear-sky emissivity, replacing two hard-coded sky temperatures	doi:10.1029/WR011i005p00742
Peng et al. (2012), <i>Environ. Sci. Technol.</i> 46(2) [A]	Equal-area rural reference method for the heat-island difference	doi:10.1021/es2030438
Voogt & Oke (2003), <i>Remote Sensing of Environment</i> 86(3) [V]	Surface vs air heat island are different quantities; view-angle sensitivity of thermal remote sensing	doi:10.1016/S0034-4257(03)00079-8
Spencer (1971), <i>Search</i> 2(5):172 [A]	Solar declination series, for per-scene sun angle	—
LP DAAC, <i>ECOSTRESS L2T LSTE v002 known issues</i> [V]	Quality-flag caveat: two documented QC bits are unset in v002	lpdaac.usgs.gov

[V] verified against full text · [A] verified from abstract or documentation only — see §6.

Full formulas and every constant: `docs/heat-map-intervention-model.md` Validation harness: `scripts/validate-model.mjs`